

Learning Activity Sheet for General Science

Quarter 1

Lesson

3

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Learning Activity Sheet for General Science Grade 11
Quarter 1: Lesson 4

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LEARNING ACTIVITY SHEET

Learning Area:	General Science	Quarter:	1
Lesson No.:	4	Date:	
Lesson Title/ Topic:	Motion Analysis Assignment		
Name:		Grade & Section:	

I. Activity No.: 1

II. Objective(s): At the end of the activity, you should be able to:

- a. identify translational and rotational motion in simple machines from daily life;
- b. analyze how forces (friction, weight, tension, torque) affect the movement and efficiency of a machine; and
- c. create labeled motion diagrams and propose improvements to enhance efficiency.

III. Materials Needed: Notebook or worksheet for observations, Smartphone or camera (optional, for documentation), Ruler or measuring tape, A selected simple machine (Bicycle, Wind-Up Toy, Scissors, Lever, etc.), and Drawing materials (paper, pencil, markers)

IV. Instructions:

Step 1: Introduction to Motion Principles

- The teacher will review the concepts of translational motion (linear movement) and rotational motion (circular movement).
- The assignment will be explained: students will observe, analyze, and diagram the motion of a simple machine found in daily life.

Step 2: Selecting a Machine and Observing Motion

- Students will choose a simple machine they frequently encounter, such as:
 - **Bicycle** – Observe the wheels (rotational motion) and forward movement (translational motion).
 - **Wind-Up Toy** – Analyze how gears rotate and drive motion in a straight line.
 - **Scissors or Lever** – Examine force application and movement efficiency.

Step 3: Data Collection and Motion Analysis

- Students will record observations, answering the following questions:

1. Which parts of the machine move in translational motion?

2. Which parts rotate, and how does rotational motion contribute to function?

3. *What forces (friction, weight, tension, torque) affect movement efficiency?*

Step 4: Diagram and Presentation Preparation

- Students will **create labeled motion diagrams** of their chosen machine.
- They will explain:
 1. *How different types of motion contribute to functionality.*
 2. *Possible improvements to enhance efficiency.*

Step 5: Class Discussion and Reflection

- Students will present their observations and findings to the class.
- The teacher will facilitate a discussion on how motion principles apply to larger machines in engineering and technology.

V. Synthesis:

1. How does motion help machines work better?

2. What did you learn about how motion improves machine designs?

3. How can engineers use motion principles to develop more efficient and sustainable machines?
